Measuring ArUco/EKF Noise on the TurtleBot3 Testbed

Experimental protocol and the tolerance bounds that follow from it

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Goal. Replace the hand-set EKF covariances and the hand-picked convergence tolerance with values measured on the testbed, using a protocol that can be repeated each session. Every bound in Section 11 is written in terms of quantities measured in Sections 5–9.

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1 What is measured and why

Symbol	Quantity	Experiment	Used for
$\sigma_x, \sigma_y, \sigma_\theta$	raw per-detection noise	A (static)	R; gate; consistency test
p_{out}	outlier (jump) fraction	A, C	gate threshold; robust estimates
$\sigma_{d,k}$	noise on edge distance k	A (static)	convergence and safety tolerances
f_{cam}, dropouts	detection rate and gaps	B (timing)	predict/update ratio; Q scaling
τ_{lat}	image-to-detection latency	B (timing)	safety margin
Q_c	process noise density	D (driving)	Q
τ_{act}	command-to-motion delay	D (driving)	safety margin; Q residual alignment
$\sigma_{e,\text{cl}}$	closed-loop edge-error jitter	E (closed-loop hold)	lowest usable `conv_tol`
$\sigma_{u\perp}$	lateral command noise	E	lower bound on lookahead l

Experiments A–D characterise the *sensor and plant*. Experiment E characterises the *closed loop*; it is the one that explains why a very small tolerance makes the formation fall apart at the end of a run (the controller ends up chasing measurement noise).

2 Models as currently implemented

Each robot i has state $\boldsymbol{x} = (x, y, \theta)$. The camera pipeline (`aruco_opencv`, then the TF from `ground_origin` to `marker_N`) gives

$$\boldsymbol{z}_k = \boldsymbol{x}_k + \boldsymbol{v}_k, \qquad \boldsymbol{v}_k \sim \mathcal{N}(\boldsymbol{0}, R), \tag{1}$$

and the predict step uses the last commanded (v, ω):

$$\boldsymbol{x}_{k+1} = f(\boldsymbol{x}_k, u_k, \Delta t) + \boldsymbol{w}_k = \begin{bmatrix} x_k + v\cos\theta_k\,\Delta t \\ y_k + v\sin\theta_k\,\Delta t \\ \theta_k + \omega\Delta t \end{bmatrix} + \boldsymbol{w}_k, \qquad \boldsymbol{w}_k \sim \mathcal{N}(\boldsymbol{0}, Q). \tag{2}$$

Facts about the code (`multi_unicycle_controller_double_angle_experiment_ekf.py`) that matter for interpreting Q and R:

1. $Q = \text{diag}(10^{-3}, 10^{-3}, 2\cdot 10^{-3})$ and $R = \text{diag}(0.05, 0.05, 0.1)$ (SI units). The theory notes (`ekf_turtlebot_theory.tex`) list a different R, $\text{diag}(0.004, 0.004, 0.008)$.
2. Q is added *per call* and is not scaled by Δt.
3. `predict()` is called at 50 Hz in the control loop *and* once in every ArUco callback, so the number of predicts between two camera updates is
$$n = 1 + \frac{50}{f_{\text{cam}}}. \tag{3}$$
4. The outlier gate rejects a detection when the squared Mahalanobis distance of the innovation exceeds $11.34 = \chi^2_3(0.99)$.

Recommended code change before measuring Q: call `predict()` once per control tick only, and use $Q(\Delta t) = Q_c\,\Delta t$ with Q_c in m^2/s and $\mathrm{rad}^2/\mathrm{s}$. Then the value measured in Experiment D transfers directly, independent of the camera rate.

3 Statistical tools used throughout

3.1 Robust spread and outliers

For a series $z_1, \ldots, z_N$ (one axis of one marker), use

$$\hat{\sigma}_{\mathrm{MAD}} = 1.4826\ \mathrm{median}_k \big| z_k - \mathrm{median}(z) \big|, \tag{4}$$

which equals σ for Gaussian data and ignores a minority of jumps. A sample is an *outlier* if $|z_k - \mathrm{median}(z)| > 5\hat{\sigma}_{\mathrm{MAD}}$ (the Gaussian probability of this is 5.7×10^{-7}, so practically every flagged sample is a genuine jump). Report the outlier fraction p_{out}, then compute the ordinary standard deviation $\hat{\sigma}$ on the inliers. If $\hat{\sigma}$ and $\hat{\sigma}_{\mathrm{MAD}}$ differ by more than about 20 %, the inliers are not Gaussian and the quantile-based bounds of Section 11 must be used instead of the Gaussian ones.

3.2 Autocorrelation and effective sample size

A Kalman filter assumes $\boldsymbol{v}_k$ is white. Compute the sample autocorrelation ρ_j of the inlier series for lags $j = 1, \ldots, J$ (stop at the first j where $|\rho_j| < 2/\sqrt{N}$). The effective number of independent samples is

$$N_{\mathrm{eff}} = \frac{N}{1 + 2\sum_{j=1}^{J} \rho_j}. \tag{5}$$

For white noise $N_{\mathrm{eff}} = N$. Report the correlation time $\tau_c = N/(N_{\mathrm{eff}} f_{\mathrm{cam}})$. If τ_c is much larger than one frame, the noise is coloured (e.g. slow lighting drift or camera vibration), and R alone does not describe it.

3.3 Confidence interval on a standard deviation

For Gaussian samples, $(N_{\mathrm{eff}} - 1)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{N_{\mathrm{eff}}-1}$, so a $1 - \alpha$ interval is

$$\hat{\sigma}\sqrt{\frac{N_{\mathrm{eff}} - 1}{\chi^2_{N_{\mathrm{eff}}-1}(1 - \alpha/2)}} \;\le\; \sigma \;\le\; \hat{\sigma}\sqrt{\frac{N_{\mathrm{eff}} - 1}{\chi^2_{N_{\mathrm{eff}}-1}(\alpha/2)}}. \tag{6}$$

For large N_{eff}, the relative half-width is about $z_{1-\alpha/2}/\sqrt{2N_{\mathrm{eff}}}$. A ± 10 % interval at 95 % therefore needs $N_{\mathrm{eff}} \approx 192$. This sets the recording length: $T \ge 192\,\tau_c$.

3.4 Are all robots the same?

To test $H_0 : \sigma_0 = \sigma_1 = \sigma_2 = \sigma_3$ (per axis), use the Brown–Forsythe test (Levene's test with medians), which is robust to non-Gaussian tails. Thin each series to every $\lceil \tau_c f_{\mathrm{cam}} \rceil$-th sample first so the samples are close to independent. If H_0 is not rejected at 5 %, pool:

$$\hat{\sigma}^2_{\mathrm{pool}} = \frac{\sum_i (N_{\mathrm{eff},i} - 1)\hat{\sigma}_i^2}{\sum_i (N_{\mathrm{eff},i} - 1)}. \tag{7}$$

If it is rejected, keep per-robot values: R is then robot-specific, which usually points to a damaged or badly mounted marker.

4 Recording setup (all experiments)

Record everything with one bag per experiment, on the machine that runs the controller:

```
ros2 bag record -o <exp>_<date> /aruco_detections /tf /tf_static \
  /bot_0/cmd_vel /bot_1/cmd_vel /bot_2/cmd_vel /bot_3/cmd_vel
```

Measurements are taken from the `ground_origin`→`marker_N` transforms, exactly as the controller sees them. The EKF output does not need to be logged separately: given the detections, the commands and the timestamps, the filter is deterministic and can be replayed offline with any Q and R. Keep the lighting, camera height and ground-origin calibration identical to the formation runs.

5 Experiment A — static noise

5.1 Design

Use four positions $P_1, \ldots, P_4$: the desired formation p^* placed at the centre of the arena. Do four 60 s sessions, rotating the robots through the positions so that each robot visits each position once (a 4×4 Latin square):

Session	P_1	P_2	P_3	P_4
1	bot_0	bot_1	bot_2	bot_3
2	bot_1	bot_2	bot_3	bot_0
3	bot_2	bot_3	bot_0	bot_1
4	bot_3	bot_0	bot_1	bot_2

This separates a robot effect (marker print, mounting) from a position effect (distance from the image centre, lens distortion, lighting). Session 1 is the actual formation geometry, so it also gives $\sigma_{d,k}$ for the real edges directly. Motors on, velocities zero (so that any vibration of a powered robot is included).

Add one extra session with a single robot visiting a 3×3 grid covering the arena, 30 s per point, to check whether σ grows towards the image edges.

5.2 Analysis

For each marker, session and axis (x, y, θ, with θ unwrapped):

1. p_{out}, $\hat{\sigma}$, $\hat{\sigma}_{\text{MAD}}$, ρ_1, N_{eff}, τ_c and the CI of $\hat{\sigma}$.
2. Robot effect: Brown–Forsythe across robots, pooled over positions. Position effect: the same test across positions, pooled over robots.
3. Cross-correlation between robots, $\rho_{ij} = \text{corr}(\delta x_i, \delta x_j)$. A large common component means camera vibration or a shared lighting effect, which cancels in relative distances (Section 11.1).
4. $\sigma_{d,k}$: the standard deviation of each logged edge distance d_k over the session, computed both from raw detections and from the replayed EKF output.
5. θ flips: report separately the fraction of samples with a heading jump larger than 20°. These come from the planar-pose ambiguity of PnP on small markers (about 40 px here) and should be treated as outliers, not folded into R_θ.

5.3 Result: R

The measured sensor covariance is

$$R_{\text{meas}} = \text{diag}(\hat{\sigma}_x^2, \hat{\sigma}_y^2, \hat{\sigma}_\theta^2) \tag{8}$$

(pooled if allowed), with the outlier fraction p_{out} reported alongside it. Off-diagonal terms can be included if the x–y correlation of a marker is significant.

6 Experiment B — timing

From the same bags:

1. **Detection rate:** for each marker, the inter-arrival times Δt_k of detections (header stamps). Report $f_{\text{cam}} = 1/\,\text{median}(\Delta t)$ and the 99th percentile of Δt.

2. **Dropouts:** the fraction of frames in which a marker is missing, and the distribution of consecutive-miss run lengths. The longest run, times $v_{\max}$, is the worst-case dead-reckoning distance.

3. **Latency:** $\tau_{\text{lat}} = t_{\text{bag}} - t_{\text{header}}$ for `/aruco_detections`, i.e. the time from image capture to the detection being available. Report the median and the 99th percentile.

7 Experiment C — jumpiness during runs

Static data cannot show motion blur or detections that jump during turns. Use the bags of normal formation runs:

1. **Innovation series.** Replay the EKF offline and store each innovation $\boldsymbol{y}_k = \boldsymbol{z}_k - \hat{\boldsymbol{x}}_{k|k-1}$ and its covariance S_k.

2. **Jump rate.** Between consecutive detections the robot moves at most $v_{\max}\Delta t_k$ (1.1 mm at 50 Hz, 2.6 mm at 21 Hz). A detection is a jump if

$$\|\boldsymbol{z}_k^{xy} - \boldsymbol{z}_{k-1}^{xy}\| > v_{\max}\Delta t_k + 5\sqrt{2}\,\sigma_{xy}, \tag{9}$$

   and similarly for θ with $\omega_{\max}$ and σ_θ. Report the jump rate per minute, per robot.

3. **Filter consistency (NIS).** The normalised innovation squared $\epsilon_k = \boldsymbol{y}_k^\top S_k^{-1} \boldsymbol{y}_k$ is χ^2_3 distributed if Q and R are correct. Over N updates,

$$\bar{\epsilon} = \frac{1}{N}\sum_k \epsilon_k \quad \text{should lie in} \quad \left[\tfrac{\chi^2_{3N}(0.025)}{N},\ \tfrac{\chi^2_{3N}(0.975)}{N}\right] \approx 3 \pm 1.96\sqrt{6/N}. \tag{10}$$

   $\bar{\epsilon} \ll 3$ means Q and/or R are larger than the real noise (an over-cautious, overly smooth filter). $\bar{\epsilon} \gg 3$ means they are too small. Use a whiteness test on $\boldsymbol{y}_k$ as well: correlated innovations mean the model is missing something, typically latency.

4. **Gate behaviour.** With correct Q, R the gate rejects $1\,\% + p_{\text{out}}$ of detections. Compare with the replayed rejection rate.

8 Experiment D — process noise and actuation delay

8.1 Design

For each robot, with `cmd_vel` recorded:

1. straight lines at $v = 0.03$ and $0.055\,\mathrm{m/s}$, 10 s each, 3 repeats;
2. turns in place at $\omega = \pm 0.13\,\mathrm{rad/s}$, 10 s each;
3. arcs at $v = 0.04\,\mathrm{m/s}$, $\omega = \pm 0.1\,\mathrm{rad/s}$;
4. step inputs, from rest to $v = 0.055$, to measure the delay.

8.2 Actuation delay

Estimate the speed $\hat{v}(t)$ from detections smoothed over about 0.3 s. τ_{act} is the shift that maximises the cross-correlation between the commanded $v(t-\tau)$ and $\hat{v}(t)$, checked against the step responses.

8.3 Estimating Q_c

For consecutive inlier detections k, $k+1$, with the commands shifted by τ_{act}, form the one-step residual

$$\boldsymbol{r}_k = \boldsymbol{z}_{k+1} - f(\boldsymbol{z}_k, u_k, \Delta t_k). \tag{11}$$

Substituting (1) and (2) and linearising f around the true state ($F_k = \partial f / \partial \boldsymbol{x}$):

$$\boldsymbol{r}_k \approx \boldsymbol{w}_k + \boldsymbol{v}_{k+1} - F_k \boldsymbol{v}_k \quad \Rightarrow \quad \mathrm{Cov}(\boldsymbol{r}_k) = Q(\Delta t_k) + R + F_k R F_k^{\top}. \tag{12}$$

Since $F_k = I + O(v\Delta t)$ and $v\Delta t \le 2.6\,\mathrm{mm}$, $F_k R F_k^{\top} \approx R$, so

$$Q(\Delta t) \approx \mathrm{Cov}(\boldsymbol{r}) - 2R_{\mathrm{meas}}. \tag{13}$$

To get the density, regress the per-axis residual variance on Δt over intervals of different lengths (use every 1st, 2nd, 4th, 8th detection): the slope is Q_c and the intercept should be $2R$, which checks Experiment A. If the slope is not distinguishable from zero, the unicycle model is better than the camera over these intervals, and Q mainly expresses how much smoothing you want (Section 10).

9 Experiment E — closed-loop hold

This experiment measures what the tolerance actually has to cover.

1. Place the robots in the desired formation (within about 2 cm).
2. Run the formation controller with the stopping rule *disabled* (set `conv_tol`= 0) for 120 s, with the variant used in the real runs.
3. Record the bag, and also log the law output $u_i = (u_x, u_y)$ and the commands (v_i, ω_i) at 50 Hz.

Report, per edge: the mean error $\bar{e}_k$, the jitter $\sigma_{e,\mathrm{cl},k}$, its correlation time, the 99.9th percentile of $|e_k|$, and whether the error drifts. For each robot, report the standard deviations of $u_{\parallel} = u \cdot h$ and $u_{\perp} = u \cdot h^{\perp}$, with $h = (\cos\theta, \sin\theta)$ and $h^{\perp} = (-\sin\theta, \cos\theta)$.

Expect $\sigma_{e,\mathrm{cl}} > \sigma_d$: near equilibrium u is mostly noise, and $\omega = u_{\perp}/l$ turns that noise into rotation, which then turns into translation through $v = 2u_{\parallel}$. If the error drifts instead of fluctuating around $\bar{e}$, the closed loop has no usable accuracy below that level, and `conv_tol` must stay above it.

10 From measured noise to filter parameters

10.1 Steady-state gain of the current filter

For one axis of a stationary robot ($v = \omega = 0$), the filter adds $q = nQ$ between updates (with n from (3)) and updates with variance r. The steady-state prior variance P^- satisfies $P^- = q + P^+$ with $P^+ = P^- r/(P^- + r)$, which gives

$$P^- = \frac{q + \sqrt{q^2 + 4qr}}{2}, \qquad K = \frac{P^-}{P^- + r} \;\approx\; \sqrt{q/r} \quad (q \ll r). \tag{14}$$

The estimate is then an exponential average of the detections. If the true detection noise is σ^2_{meas} (whatever value is used for r), the noise left in the estimate is

$$\sigma^2_{\text{est}} = \sum_{j \geq 0} K^2 (1-K)^{2j} \sigma^2_{\text{meas}} = \frac{K}{2-K}\, \sigma^2_{\text{meas}}, \tag{15}$$

and the averaging time constant is $\tau_f \approx 1/(K f_{\text{cam}})$.

Current values, with $f_{\text{cam}} \approx 21\,\text{Hz}$ (so $n \approx 3.4$): position $q = 3.4 \times 10^{-3}$, $r = 0.05$ give $K = 0.23$, and heading $q = 6.8 \times 10^{-3}$, $r = 0.1$ also give $K = 0.23$, because Q/R is 0.02 on both axes. Hence $\tau_f \approx 0.21\,\text{s}$ and $\sigma_{\text{est}} \approx 0.36\,\sigma_{\text{meas}}$.

10.2 Two defensible choices

1. **Consistent filter.** Set $R = R_{\text{meas}}$ and $Q = Q_c \Delta t$ from Experiments A and D. Justification: both are measured. Check: $\bar{\epsilon}$ (Experiment C) inside its interval.

2. **Deliberately smoothed filter.** Only the ratio Q/R sets K and therefore the estimate; the absolute scale sets only P and the gate. Choose K from the smoothing time you want, $K = 1/(\tau_f f_{\text{cam}})$, then $q/r = K^2/(1-K)$ (from (14)). Justification: report R_{meas}, the chosen τ_f, and why that amount of smoothing is needed (for example jumps, latency, or ω noise in Experiment E). Then scale R so that the gate still tests against the measured noise, i.e. the NIS check stays meaningful.

Using R several hundred times R_{meas} (as now) without stating which choice was made cannot be justified as a noise model, only as smoothing.

10.3 Gate threshold

Along one axis, the gate admits innovations up to $|y| \leq \sqrt{11.34\,S}$ with $S = P^- + r$. With the current values this is 0.86 m in position and 69° in heading, so PnP heading flips pass through. With $R = R_{\text{meas}}$, a 5σ jump is rejected. Choose the threshold from the jump-size distribution of Experiment C: it should separate the genuine jumps from the Gaussian part.

11 Tolerance bounds

11.1 Noise on an edge distance

For $d = \|p_i - p_j\|$ with small position errors $\delta p_i, \delta p_j$ and unit vector $\hat{u} = (p_i - p_j)/d$:

$$\delta d \approx \hat{u}^\top (\delta p_i - \delta p_j) \;\Rightarrow\; \sigma_d^2 = \hat{u}^\top \big(\Sigma_i + \Sigma_j - \Sigma_{ij} - \Sigma_{ji}\big)\hat{u}, \tag{16}$$

where Σ_{ij} is the cross-covariance. With independent isotropic errors this is $\sigma_d = \sqrt{2}\,\sigma_p$. Correlated errors (camera vibration) cancel. Use the directly measured $\sigma_{d,k}$ from Experiment A, and (16) only as a cross-check. Note that σ_p here is the noise of the quantity the controller uses: the EKF estimate, σ_{est} from (15), not the raw detection.

11.2 Convergence tolerance

Let $e_k^m = e_k + n_k$ be the measured error of edge k, with jitter σ. While the robots are moving, $\sigma = \sigma_{e,\mathrm{cl}}$ from Experiment E. While they are stopped, $\sigma = \sigma_d$.

Lower limit on the stop threshold. For the stop condition $|e_k^m| \le \texttt{tol}$ on all 5 edges to be reached and held without the controller chasing noise, the threshold must clear the closed-loop jitter. Using Bonferroni over 5 edges with total probability α:

$$\texttt{tol}_{\mathrm{in}} \;\ge\; |\bar{e}_k| + z_{1-\alpha/10}\,\sigma_{e,\mathrm{cl},k} \quad \text{for all } k, \qquad z_{1-0.001} = 3.09 \;(\alpha = 0.01). \tag{17}$$

If the jitter is not Gaussian, use the 99.9th percentile of $|e_k|$ from Experiment E directly. This is the quantitative form of the observation that a very small `conv_tol` breaks the formation at the end of a run.

Resume threshold (hysteresis). Once stopped, a false restart happens if noise pushes a measured error from $\texttt{tol}_{\mathrm{in}}$ above $\texttt{tol}_{\mathrm{out}} = \texttt{tol}_{\mathrm{in}} + h$. In the worst case the true error sits exactly at $\texttt{tol}_{\mathrm{in}}$. Over a hold of T seconds there are $N_{\mathrm{eff}} = T/\tau_c$ independent samples per edge. For a total false-restart probability α over 5 edges,

$$h \;\ge\; \sigma_d\,\Phi^{-1}\!\Big(1 - \frac{\alpha}{5N_{\mathrm{eff}}}\Big). \tag{18}$$

Example: $T = 60\,\mathrm{s}$, $\tau_c = 0.2\,\mathrm{s}$, $\alpha = 0.05$ gives $N_{\mathrm{eff}} = 300$ and $h \ge 3.99\,\sigma_d$.

Hold time before stopping. If the stop condition must hold for M consecutive independent samples, a configuration whose true error is δ above $\texttt{tol}_{\mathrm{in}}$ passes by chance with probability at most $\Phi(-\delta/\sigma)^M$. Given σ and M, this defines how far above $\texttt{tol}_{\mathrm{in}}$ the true error can still be when the robots stop.

Accuracy that can be claimed. After stopping, the true error satisfies, with confidence $1-\alpha$ over 5 edges,

$$|e_k| \le \texttt{tol}_{\mathrm{in}} + z_{1-\alpha/10}\,\sigma_d/\sqrt{M_{\mathrm{eff}}}, \tag{19}$$

where M_{eff} is the number of independent samples averaged in the stop check (1 for the raw check). A final error inside $\texttt{tol}_{\mathrm{in}}$ is therefore the dead band of the stopping rule, not a steady-state error of the control law.

11.3 Safety and connectivity margins

The controller is meant to keep $0.295\,\mathrm{m} \le d \le 1.465\,\mathrm{m}$ between centres. The estimated distance lags the true one by the latency and the filter delay, and it is noisy. With the maximum closing speed $2v_{\max}$:

$$d_{\mathrm{true}} \;\ge\; d_{\mathrm{meas}} - z\,\sigma_d - 2v_{\max}\,(\tau_{\mathrm{lat}} + \tau_{\mathrm{act}} + \tau_f). \tag{20}$$

So a logged distance proves the lower bound held only if $d_{\mathrm{meas}} \ge 0.295 + z\sigma_d + 2v_{\max}(\tau_{\mathrm{lat}} + \tau_{\mathrm{act}} + \tau_f)$, and symmetrically for the upper bound. Example: $\sigma_d = 10\,\mathrm{mm}$, $z = 3$, and a total delay of $0.4\,\mathrm{s}$ give a margin of $30 + 44 = 74\,\mathrm{mm}$. Use the same margin when calling a logged dip below $0.295\,\mathrm{m}$ a real violation.

11.4 Lower bound on the lookahead length

From the conversion $\omega = u_\perp / l$, noise in the law output becomes turn-rate noise $\sigma_\omega = \sigma_{u\perp}/l$. Heading noise contributes $\delta u_\perp \approx -u_\parallel \delta\theta$, which is negligible near convergence where u is small. Requiring the turn-rate noise to use at most a fraction β of $\omega_{\max}$:

$$l \geq \frac{\sigma_{u\perp}}{\beta\, \omega_{\max}}. \tag{21}$$

With $\sigma_{u\perp}$ from Experiment E, this states the reason for $l = 0.25\,\mathrm{m}$ in measured terms. With $\beta = 0.25$ and $\omega_{\max} = 0.13\,\mathrm{rad/s}$, the current l is justified if $\sigma_{u\perp} \leq 8\,\mathrm{mm/s}$.

12 Results sheet

Quantity	Value (with 95 % CI)	Exp.
σ_x, σ_y (pooled? Brown–Forsythe p)		A
σ_θ, θ-flip fraction		A
p_{out}, τ_c		A
$\sigma_{d,k}$ for $k = 1..5$ (raw / EKF)		A
f_{cam}, dropout fraction, longest gap		B
τ_{lat} (median / p99)		B
jump rate, $\bar{\epsilon}$ (NIS), gate rejection rate		C
Q_c (position / heading), τ_{act}		D
$\bar{e}_k$, $\sigma_{e,\text{cl},k}$, p99.9 $\lvert e_k \rvert$, $\sigma_{u\perp}$		E
$\texttt{tol}_{\text{in}}$ (17), h (18)		–
safety margin, minimum l		–

A Evidence available before these experiments

These numbers come from data that was not recorded for this purpose. They only bracket the answer.

Source	Value	Limitation
Case-4b overhead video, marker centroid and edge angle, 1 s detrended	$\sigma_{xy} \approx 0.55\,\mathrm{mm}$, $\sigma_\theta \approx 1.1°$, no flips	2D image measurement, not the PnP pose the EKF uses; MP4 compression smooths the noise; a **lower bound**
Case-4b EKF log, robots mostly stopped, 80–150 s	$\sigma_{d,k} = 8$–$17\,\mathrm{mm}$, peak-to-peak up to 88 mm	includes short restarts of the controller, so it is partly real motion; an **upper bound**
Case-4b video frame count vs. run length	$f_{\text{cam}} \approx 21\,\mathrm{Hz}$	the recorder's rate, assumed to match the detector's